

			$M_B = 2.0 \times 10^{-3} \text{ kg}$	1
6	(a)	(i)	Upthrust and Weight	1
		(ii)	<u>At equilibrium, the magnitude of upthrust and weight are equal. When the tube is pushed downwards, the magnitude of upthrust increases.</u> Since the weight of the tube remains constant, the <u>resultant force is upwards</u>	1 1
	(b)		Since $\frac{A\rho g}{M}$ is a constant, acceleration of the tube is proportional to its displacement The negative sign indicates that acceleration and displacement are in the opposite direction	1 1
	(c)	(i)	Period of the oscillation = 2.2 s Angular frequency = $2\pi/T$ $= 2\pi/2.2$ $= 2.9 \text{ rad s}^{-1}$	1

		(ii)	$\omega^2 = \frac{A\rho g}{M}$ $2.9^2 = \frac{(4.5 \times 10^{-4})\rho(9.81)}{(0.17)}$ $\rho = 323.86 = 320 \text{ kg m}^{-3}$	1 1
		(iii)	$v_0 = \omega x_0$ $v_0 = (2.9)(0.03)$ $= 0.087 \text{ m s}^{-1}$	1 1

7	(a)	(i)	W	1
---	-----	-----	-----	---